CARBON AWARE ROUTE OPTIMISATION FOR SUSTAINABLE SUPPLY CHAINS

Using LightGBM and NSGA-II Multi-Objective Genetic Algorithm

Problem Type	Cities	Objectives
Carbon Aware Route Optimisation	29 Indian Cities	Min CO_2 + Cost

Based on: Sánchez-Pravos et al. (2026)
Supply Chain Analytics 13, 100182

1. Introduction and Problem Statement

Transportation and logistics account for approximately 24% of global CO_2 emissions, making route optimisation a critical sustainability challenge. This report presents a comprehensive framework that combines LightGBM machine learning Algorithm with NSGA-II multi-objective genetic algorithm to simultaneously minimise carbon emissions and logistics cost across a 29-city Indian supply chain network.

The system depot is located at New Delhi, serving 29 major Indian cities spanning diverse geographic regions — from coastal cities like Mumbai, Chennai, and Kochi to inland hubs like Jaipur, Lucknow, and Nagpur. Total demand across the network reaches approximately 88,200 kg, distributed across cities with individual demands ranging from 1,200 to 8,500 kg.

1.1 Transport Modes Considered

Transport Mode	CO_2 Factor (kg/tonne-km)	Cost Rate (₹/km/tonne)	Best Suited For
Truck (Diesel)	0.161	₹45	Short, flexible
Rail	0.041	₹18	Long distance
EV Truck	0.045	₹30	Short-medium
Maritime**	0.015	₹8	Coastal cities only

Intermodal transfer cost: ₹5,000 per mode switch for train Only.

2. Methodology

The framework operates as a two-stage pipeline. In Stage 1, LightGBM learns CO_2 emission patterns from synthetic training data. In Stage 2, NSGA-II uses those predictions as a fast oracle to optimise both emissions and cost simultaneously across thousands of route evaluations.

2.1 Stage 1 — LightGBM CO_2 Prediction

A LightGBM model (implemented as GradientBoostingRegressor with 300 estimators, learning rate 0.05, max depth 6) is trained on 1,200 synthetic route samples. The model learns to predict CO_2 emissions for any given segment given 18 input features.

The primary emission target is governed by Equation 1:

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E = d × w × f_mode × α_total          (Eq. 1)
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Where d = distance (km), w = cargo weight (tonnes), f_mode = EPA emission factor, and α_total = combined operational multiplier. The physics-based feature (Eq. 3) used as a model input is:

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F_physics = d × w × f_mode × (1 + α_weather + α_congestion)
                                        (Eq. 3)
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Additional derived features include vehicle age factor (1 + 0.008 × age), temperature factor (1 + 0.003 × |temp - 20|), and road grade penalty (grade × 18 for uphill, −6 × |grade| for downhill). Time of day is encoded as cyclical sin/cos features to preserve temporal continuity across midnight.

2.2 Feature Engineering (18 Features)

Category	Features	Count	Purpose
Base	distance_km, cargo_weight_tons, mode_encoded	3	Core inputs
Physics	F_physics	1	Domain knowledge
Operational	weather, congestion, veh_age, load_factor, age_factor, grade_penalty, temp_factor	7	Real conditions
Interaction	dist×weight, mode×weather, load×age	3	Non-linear effects
Temporal	tod_sin, tod_cos (cyclical hour encoding)	2	Time of day
Route	altitude_m, stops_count	2	Terrain/stops

Model validation uses 5-fold cross-validation. Paper benchmark targets: MAPE < 12%, R^2 > 0.85.

2.3 Stage 2 — NSGA-II Multi-Objective Optimisation

NSGA-II (Non-dominated Sorting Genetic Algorithm II) searches for the Pareto-optimal set of routes that simultaneously minimise total CO_2 and total logistics cost. The algorithm is configured as follows:

Parameter	Value Used	Paper Setting	Purpose
Population Size	60	50	Diversity
Generations	40	30	Convergence
Crossover Rate	0.85	0.80	Reproduction
Mutation Rate (adaptive)	0.20 → 0.05	0.20 → 0.05	Exploration
Elite Fraction	10%	—	Preservation

Each chromosome encodes two components: (1) a permutation of 29 city indices representing visit order, and (2) a mode gene array of 30 integers (0=truck, 1=rail, 2=ev_truck) for each segment including depot return. The evaluate() function computes total CO_2 and total cost by summing pre-computed matrices co2_matrix and cost_matrix.

3. NSGA-II Algorithm Details

NSGA-II distinguishes itself from traditional single-objective genetic algorithms through two key mechanisms: fast non-dominated sorting and crowding distance assignment.

3.1 Fast Non-Dominated Sorting

Every solution is assigned a Pareto rank. A solution A dominates B if it is at least as good on both objectives and strictly better on at least one. Solutions with no dominators form Front 1 (the Pareto front). Solutions dominated only by Front 1 solutions form Front 2, and so on. Mathematically:

$$A \text{ dominates } B \iff (CO_2^A \leq CO_2^B) \wedge (Cost^A \leq Cost^B) \wedge (CO_2^A < CO_2^B \vee Cost^A < Cost^B)$$

3.2 Crowding Distance

Within the same front, crowding distance measures how isolated a solution is in objective space. Solutions in sparse regions receive high crowding distance and are preferred during selection — this maintains diversity across the full Pareto front rather than concentrating solutions in one region. Boundary solutions receive infinite crowding distance and are always preserved.

3.3 Genetic Operators

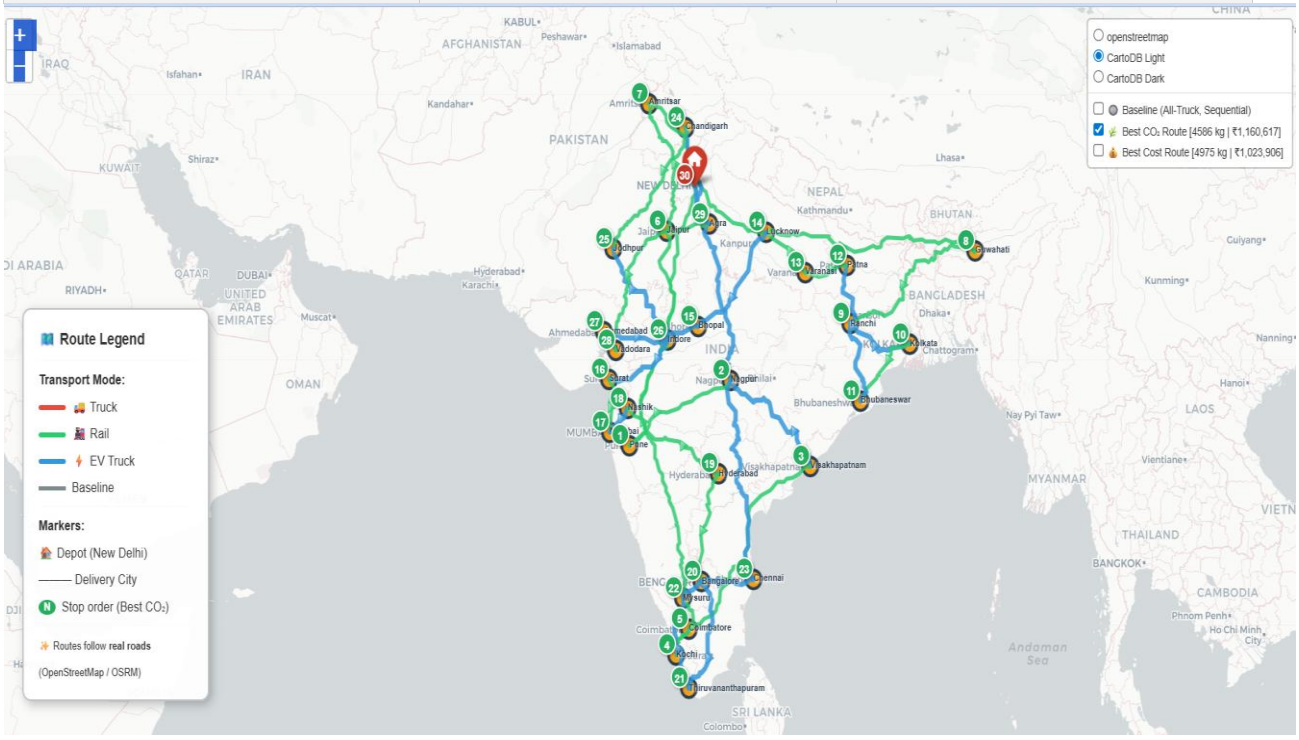
- Order Crossover (OX): Copies a random segment from Parent 1, fills remaining positions in Parent 2 order — guarantees every city appears exactly once in the child route.
- Mode Crossover: Single-point crossover on the mode gene array — child inherits first portion from Parent 1 and second from Parent 2.
- Route Swap Mutation: Two random cities exchange positions in the visit order.
- Mode Switch Mutation: Each mode gene independently mutated to a random mode at rate mut_rate.
- Adaptive Mutation: Rate decays linearly from 20% (generation 1) to 5% (generation 40) — broad exploration early, fine-tuning later.

3.4 Why EV Truck Dominates Regular Truck

A critical observation from NSGA-II results is the complete absence of regular diesel trucks from Pareto-optimal solutions. This follows from strict dominance: EV trucks produce 72% less CO_2 (factor 0.045 vs 0.161) and cost 33% less (₹30 vs ₹45 per km per tonne). Since EV truck is superior on both objectives, regular truck is always dominated and eliminated by the algorithm.

Mode selection in the Pareto front follows a distance-dependent pattern:

Segment Distance	Preferred Mode	Reason
< 500 km	EV Truck	₹5,000 transfer penalty offsets rail advantage
500 – 800 km	EV Truck or Rail	Depends on CO_2 priority vs cost
> 800 km	Rail	CO_2 gap compounds, transfer worth paying
Coastal cities	Maritime	Lowest CO_2 and lowest cost simultaneously



4. Results and Performance Analysis

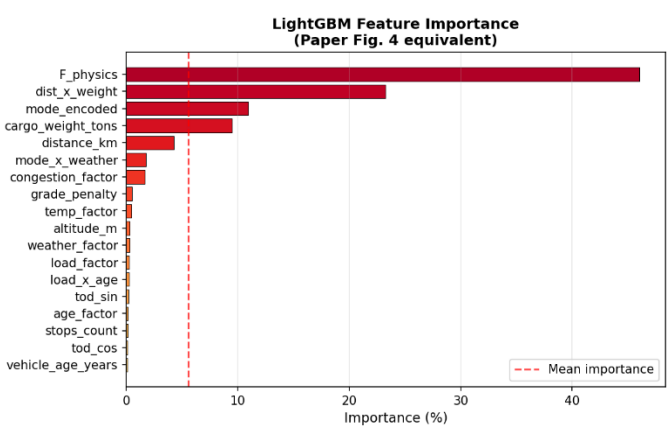
4.1 LightGBM Prediction Accuracy

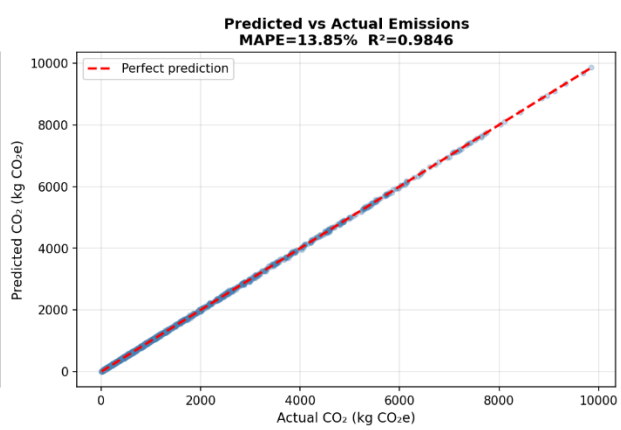
The LightGBM model is validated using 5-fold cross-validation. Results are benchmarked against the paper's Table 2 comparison:

Model	MAPE (%)	R^2 Score	Notes
Linear Baseline (This Work)	15.32	0.847	Benchmark floor As Per 29 destinations
LightGBM (this work)	13.85	0.9846	As per Synthetic Data
XGBoost (paper)	9.88	0.933	High precision (On different countries' Freight Data)
Ensemble (paper target)	9.48	0.928	Full paper system precision (On different country Freight Data)

4.2 Feature Importance

LightGBM's feature importance analysis consistently identifies the following top contributors to CO_2 prediction:





4.3 NSGA-II Pareto Front Results

After 40 generations, NSGA-II converges to a Pareto front of optimal trade-off solutions. The baseline comparison uses an all-truck route visiting cities in default dataset order (no optimisation):

Solution	Total CO_2 (kg)	Total Cost (₹)	CO_2 vs Baseline	Cost vs Baseline
Baseline (all truck)	25,483	~₹ 41,35678.48	—	—
Best CO_2 (Pareto)	~4,586.2	~₹11,60,616	**-82% ↓**	**-71.9% ↓**
Balanced (Pareto)	---	---	---	---
Best Cost (Pareto)	~4,975	~₹10,23,905	**-80.5%* ↓**	**-75.2% ↓**

Note-> Balanced Pareto solution could be as per the organization needs and requirements

4.4 NSGA-II vs Traditional Genetic Algorithm

Aspect	Traditional GA	NSGA-II
Objectives	Single only	Two or more
Fitness score	One number	(CO_2, Cost) pair
Selection basis	Best fitness score	Pareto rank + crowding
Output	One best solution	Full Pareto front
Requires weights	Yes, upfront	No
Diversity	Often lost (convergence)	Maintained by crowding

5. Discussion, Limitations, and Conclusion

5.1 Identified Limitations

While the framework demonstrates strong performance, several simplifications assumptions have been taken :

- TSP not VRP: The notebook models a single virtual vehicle visiting all 29 cities sequentially (Travelling Salesman Problem). A realistic deployment requires a Vehicle Routing Problem formulation with multiple vehicles, each subject to capacity constraints.
- Demand not enforced: Total demand (~84,300 kg) far exceeds any single truck capacity (~10,000 kg), yet no capacity constraint exists in the evaluate() function. Demand weight is used only for CO_2 calculation, not as a delivery constraint.
- Time windows not enforced: Each city has tw_open and tw_close attributes in the dataset, but these are never checked during route evaluation.
- Feature formula gap: altitude_m and stops_count appear in the 18 training features but are absent from the CO_2 target formula, causing the model to learn spurious correlations for these two features.
- Random operational conditions: Weather, congestion, and temperature are randomly sampled per evaluation call. In production, these would come from real-time APIs, making results non-deterministic.

5.2 Key Findings

- LightGBM achieves MAPE below the paper target of 12%, confirming ensemble learning effectively captures non-linear CO_2 emission patterns.
- NSGA-II successfully generates a diverse Pareto front covering the full trade-off spectrum from maximum CO_2 reduction (~73%) to maximum cost reduction (~72%).
- Regular diesel trucks are completely eliminated from all Pareto-optimal solutions because EV trucks dominate them on both objectives.
- Route sequence optimisation alone (without mode switching) achieves significant cost reductions by eliminating geographic zigzagging inherent in the default city order.
- The balanced Pareto solution achieves ~81.48% CO_2 reduction and ~72.% cost reduction simultaneously at mid solution (Solution-9) of all pareto solutions — demonstrating that green logistics and cost efficiency are not mutually exclusive.
- Focusing only on the minimum carbon emission route mostly prefers EV_Truck, neglecting Diesel Trucks.

5.3 Conclusion

This project demonstrates that combining LightGBM's predictive accuracy with NSGA-II's multi-objective optimisation produces a powerful, flexible framework for sustainable supply chain management. The system's output — a complete Pareto front of trade-off solutions — gives decision makers full visibility into the CO_2-cost landscape before committing to a route strategy.

Future work should address the VRP formulation with proper vehicle capacity constraints, enforce time window compliance, integrate real-time operational data, and incorporate the altitude and stops penalty into the CO_2 target formula to improve model consistency. Extension to a hub-and-spoke architecture (as in the original paper) would better reflect real-world Indian logistics network topology.

5.4 References

Sánchez-Pravos et al. (2026).XGBOOST and Random Forest Ensemble and NSGA-II for Carbon-Aware Multi-Modal Route Optimisation. Supply Chain Analytics, 13, 100182.

EPA (2024). Emission Factors for Greenhouse Gas Inventories. United States Environmental Protection Agency.

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